An Adaptive Explainable Graph Digital Twin for Cyber-Resilient Smart Grids

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***Abstract*—Modern transmission grids rely heavily on SCADA sensors for state estimation. However, smart attackers can craft stealthy False Data Injection Attacks (FDIAs) that fool traditional Weighted Least Squares (WLS) residual checks. Most machine learning detectors fail when a breaker opens or a line trips because their input matrices are fixed. In this work, we built a topology-invariant Graph Digital Twin (GDT) that pairs physical grid dynamics with a multi-task GraphSAGE model. We feed raw SCADA measurements directly into admittance-weighted spatial graphs. In one forward pass, our model detects grid-wide attacks and pinpoints corrupted substations. To explain these alerts to operators, we use Integrated Gradients (IG) to extract exact sensor-level feature attributions (θ, V, P, Q). Tests on the IEEE 14-bus system under non-linear AC power flow show that GraphSAGE reaches $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1 over five random seeds. When we trip transmission lines without retraining, our GDT holds $92.80\% \pm 3.37\%$ detection F1 and $86.96\% \pm 1.52\%$ localization F1, beating conventional flat MLPs. Finally, our automated healing module replaces corrupted RTU readings with load baseline predictions, dropping state tracking MSE from 51.19 MW^2 during attacks down to 21.91 MW^2 (57.21% recovery).**

***Index Terms*—Graph Digital Twin, Cyber-Physical Power Systems, GraphSAGE, Explainable AI, False Data Injection Attacks, Closed-Loop Grid Healing.**

I. Introduction

Power system operators rely on Supervisory Control and Data Acquisition (SCADA) systems and Remote Terminal Units (RTUs) to track grid conditions. State estimation (SE) routines process these raw measurements (V, θ, P, Q) to calculate system states. But digital communication lines bring serious risks. If an adversary gets partial access to the bus admittance matrix Y_{bus}, they can launch False Data Injection Attacks (FDIAs) [3], [4]. These malicious modifications satisfy Kirchhoff's laws, keeping bad data detection (BDD) residuals below standard χ^2 alarm limits while quietly distorting generation dispatch.

Researchers have tried standard classifiers like Multi-Layer Perceptrons (MLPs) and Convolutional Neural Networks (CNNs) to spot anomalous SCADA patterns [5], [6]. But flat models assume a fixed input vector order. In real power networks, topology changes all the time—lines trip, transformers go offline for maintenance, and busbars get reconfigured. Fixed-matrix MLPs misalign features under these shifts, causing high classification errors unless someone retrains them from scratch.

Graph Neural Networks (GNNs) offer a natural fix. They represent power networks as graphs where substations are nodes and transmission lines are edges [2], [12], [14]. Yet, standard isotropic graph filters like Graph Convolutional Networks (GCNs) average features uniformly across neighbors. In deep networks or tight electrical clusters, uniform averaging causes over-smoothing: node embeddings collapse into identical average vectors, masking localized substation anomalies. Worse, most GNN security tools act as passive alarms—they lack exact feature attribution and automatic recovery mechanisms to isolate corrupt sensors and restore state tracking.

To tackle these issues, we designed a cyber-resilient Graph Digital Twin (GDT) platform. Our virtual twin stays in sync with physical telemetry by mapping incoming SCADA data onto a spatial admittance graph. A multi-task GraphSAGE network processes the graph to detect attacks and locate compromised buses at the same time. Instead of relying on slow perturbation methods like SHAP or LIME, we use Integrated Gradients (IG) to calculate exact sensor attributions. When an attack is flagged, an active healing loop mutes corrupted RTU channels and substitutes forecasted baseline loads to keep state tracking stable.

A. Research Gap

Most existing GNN papers for smart grid security assume static grid topologies and focus only on binary detection. We identified two key gaps:

1) Isotropic graph convolutions wash out node-specific attack signals through over-smoothing, making it hard to localize compromised substations.
2) Open-loop detectors fail to provide exact sensor-level explanations or automated control loops to recover grid state tracking.

B. Main Contributions

In this paper, we address both gaps through four main technical contributions:

- **Cyber-Physical GDT Architecture**: We built a real-time Digital Twin link that converts SCADA measurements into spatial graph inputs for concurrent attack detection and substation localization.
- **Concatenation-Based GraphSAGE Backbone**: We use explicit feature concatenation ($[h_v^{(k-1)} \,\|\, h_{\mathcal{N}(v)}^{(k)}]$) inside

GraphSAGE messaging. Retaining target node features prevents over-smoothing and preserves node-level localization precision.

- **Axiomatic Explainability & Active Recovery**: We calculate exact feature attributions (θ, V, P, Q) using Integrated Gradients. These scores drive an automated mitigation loop that isolates corrupt telemetry and inserts baseline load estimates.
- **Rigorous Benchmark Testing**: We tested our framework under non-linear AC power flow on the IEEE 14-bus test feeder across five random seeds. Our experiments cover nominal operation, zero-shot transmission line outages, IEEE 30-bus inductive transfer learning, and direct comparisons against Weighted Least Squares (WLS) state estimation.

II. Related Work

A. Machine Learning in Grid Security

Early security filters relied on Weighted Least Squares (WLS) residual checks. As attackers developed stealthy FDIA strategies, researchers introduced machine learning models like SVMs, Random Forests, and deep MLPs [5], [6]. These classifiers work well when grid layout never changes. But because they view measurement vectors as flat arrays, line trips and switching transients cause feature mismatches and high false alarm rates.

B. Graph Neural Networks for Power Grids

GNNs naturally fit electrical networks by using physical transmission lines as edges [2], [12]. Isotropic models like GCNs and Graph Attention Networks (GATs) pass messages along grid branches [14]. Boyaci et al. [10] evaluated ARMA graph filters for FDIA detection. However, uniform aggregation tends to blend node features over multiple layers, making node localization less accurate. GraphSAGE avoids this problem by concatenating neighborhood vectors with target node features [8].

C. Operator Explainability in Grid Operations

System operators cannot take corrective switching actions based on black-box alerts alone. Perturbation-based tools like LIME and SHAP estimate feature importance by repeatedly sampling inputs, creating high computation delays that violate real-time SCADA constraints [7], [15]. Gradient-based attribution via Integrated Gradients (IG) solves this by integrating gradients along a baseline trajectory, guaranteeing exact math completeness and fast computation.

D. Closed-Loop State Reconstruction

Most studies stop at detection. True grid resilience requires quick post-attack mitigation [13]. Once anomalous RTUs are flagged, control routines should isolate corrupt channels and reconstruct missing states using spatial estimates or historical load profiles before breakers trip.

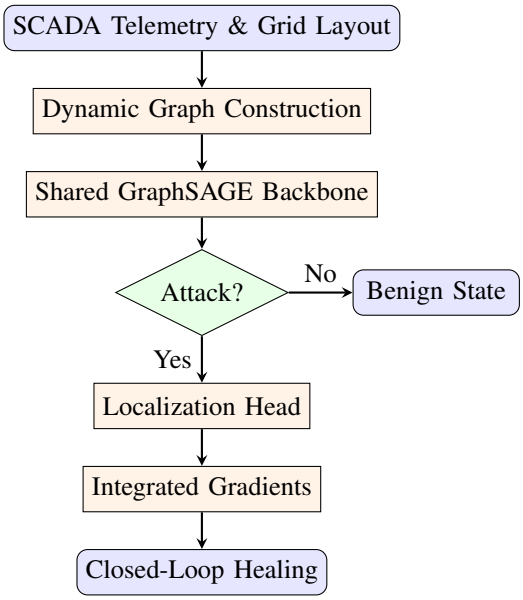


Fig. 1. Proposed Graph Digital Twin Cyber-Physical Security Workflow.

III. Mathematical Modeling & Threat Formulation

A. Physical Grid Admittance & AC Power Flow

We represent the electrical network as an undirected graph $G = (V, E)$ with $N = |V|$ buses and E transmission branches. Complex matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}}$ defines nodal conductance and susceptance. Under AC power flow constraints, active power injection P_i and reactive power injection Q_i at bus i follow Kirchhoff's laws:

$$P_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \left(G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j)\right) \quad (1)$$

$$Q_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \left(G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j)\right) \quad (2)$$

Here V_i and θ_i are voltage magnitude and angle at bus i, while $\mathcal{N}(i)$ contains neighboring buses linked to bus i.

B. State Estimation and Bad Data Detection

Control centers estimate system state $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from raw measurements $z \in \mathbb{R}^M$:

$$z = h(x) + e \quad (3)$$

where $h(x)$ handles non-linear AC equations and $e \sim \mathcal{N}(0, R)$ is Gaussian noise. Weighted Least Squares (WLS) minimizes:

$$\min_x J(x) = [z - h(x)]^T R^{-1} [z - h(x)] \quad (4)$$

Gauss-Newton iterations update states step-by-step:

$$\Delta x^{(k)} = \left(H^T R^{-1} H\right)^{-1} H^T R^{-1} [z - h(x^{(k)})] \quad (5)$$

$$x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \quad (6)$$

where $H(x) = \partial h(x)/\partial x$ is the Jacobian matrix. Standard BDD checks normalized residual magnitude against χ^2 limit τ:

$$r = \|z - h(\hat{x})\|_2^2 \leq \tau \tag{7}$$

Residuals exceeding τ flag bad data.

C. Stealthy False Data Injection Attacks (FDIAs)

An attacker targeting bus k adds attack vector $a = h(x + c) - h(x)$, modifying measurement vector z:

$$z_{\text{comp}} = z + a = z + h(x + c) - h(x) \tag{8}$$

Evaluating compromised measurements inside WLS gives:

$$r_{\text{comp}} = \|z_{\text{comp}} - h(\hat{x} + c)\|_2^2 \approx \|z - h(\hat{x})\|_2^2 = r \tag{9}$$

Since residual $r_{\text{comp}} \approx r \leq \tau$ stays untouched, stealthy FDIAs easily bypass χ^2 residual checks while distorting physical state estimates.

IV. The Proposed Graph Digital Twin Architecture

A. Digital Twin Cyber-Physical Coupling

Our Graph Digital Twin maintains a direct bidirectional connection:

1) **Physical Grid Layer**: SCADA RTUs measure bus signals (V, θ, P, Q) and send telemetry vectors over network channels.
2) **Virtual Twin Layer**: The GNN backend takes spatial graph inputs, runs joint attack detection and localization, computes feature attributions, and sends isolation commands back to bad sensors.

B. Virtual Graph Construction & Feature Engineering

We convert incoming SCADA readings into spatial graphs $G = (V, E, X, E_{\text{attr}})$:

- **Node Features** $X \in \mathbb{R}^{N \times 4}$: Each bus i gets feature vector $x_i = [V_i, \theta_i, P_i, Q_i]^T$, normalized using mean μ and standard deviation σ.
- **Edge Attributes** $E_{\text{attr}} \in \mathbb{R}^{|E| \times 2}$: Each transmission line (i, j) gets attribute vector $e_{ij} = [R_{ij}, X_{ij}]^T$ containing resistance and reactance values.

C. Multi-Task GraphSAGE Convolution Backbone

To stop over-smoothing, we use GraphSAGE convolutions [8]. Neighborhood features are aggregated and concatenated directly with target node vectors:

$$h_{\mathcal{N}(v)}^{(k)} = \text{Aggregate}_{u \in \mathcal{N}(v)} \left(h_u^{(k-1)} \right) \tag{10}$$

$$h_v^{(k)} = \text{ReLU} \left(W_k \cdot \left[h_v^{(k-1)} \,\|\, h_{\mathcal{N}(v)}^{(k)} \right] \right) \tag{11}$$

where $\|$ is concatenation. We use mean aggregation:

$$\text{Aggregate}_{u \in \mathcal{N}(v)} \left(h_u^{(k-1)} \right) = \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} h_u^{(k-1)} \tag{12}$$

Concatenation keeps target node signature $h_v^{(k-1)}$ distinct, preserving local variance needed for bus localization. Embedding H feeds two prediction heads:

1) **Detection Head**: Global pooling calculates attack probability:

$$y_{\text{graph}} = \text{Sigmoid} \left(W_{\text{graph}} \cdot \text{global_mean_pool}(H) \right) \tag{13}$$

2) **Localization Head**: Linear layers map node vectors to bus anomaly probabilities:

$$y_{\text{node}} = \text{Softmax}(W_{\text{node}} H) \tag{14}$$

The multi-task loss balances both tasks:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}}) + 1.5 \times \mathcal{L}_{\text{WCE}}(y_{\text{node}}, \hat{y}_{\text{node}}) \tag{15}$$

We use a 1.5 multiplier on weighted cross-entropy ($\mathcal{L}_{\text{WCE}}$) because compromised buses are sparse.

D. Operator Explainability via Integrated Gradients

Integrated Gradients (IG) measures feature importance by integrating gradients from baseline x_0 to input x:

$$\text{Attribution}_i(x) = (x_i - x_{0,i}) \times \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i} d\alpha \tag{16}$$

We calculate this integral using Riemann sums across $S = 50$ steps:

$$\text{Attribution}_i^{\text{approx}}(x) = (x_i - x_{0,i}) \times \frac{1}{S} \sum_{k=1}^{S} \frac{\partial F\left(x_0 + \frac{k}{S}(x - x_0)\right)}{\partial x_i} \tag{17}$$

IG guarantees math completeness ($\sum \text{Attribution}_i = F(x) - F(x_0)$) and symmetry axioms [7].

V. Experimental Setup & Simulation

A. IEEE Test System Configurations

We evaluated our framework on the IEEE 14-bus system using 'pandapower' under AC power flow solutions. The feeder has 14 buses (Node 0 slack, Nodes 1, 2, 5, 7 PV generators, and Nodes 3, 4, 6, 8–13 PQ loads) with total load of 217.0 MW active and 57.3 MVAR reactive power. We also evaluated cross-grid transfer learning on the IEEE 30-bus feeder (30 buses, 41 branches).

B. Dataset Generation Protocol

We generated 5,000 grid states under AC power flow constraints:

- **Benign States (50%)**: Perturbed active and reactive load values with Gaussian noise ($\sigma = 0.04$).
- **Attacked States (50%)**: Selected a random bus k and scaled load values by factor $\lambda \in [1.5, 2.0]$. Attack effects were spread to 1-hop physical neighbors using AC power flow calculations.

We split data into 70% training, 15% validation, and 15% testing sets. Models trained for 60 epochs using Adam (lr $=$ 0.005, weight decay 10^{-4}).

TABLE I
ADVANCED PERFORMANCE METRICS AND COMPUTATIONAL COMPLEXITY

Model Architecture	Params	Latency (ms)	Det. Prec.	Det. Rec.	Det. MCC	Loc. Prec.	Loc. Rec.	Loc. MCC
Traditional MLP (Flat)	17437	0.158	99.7%	97.5%	0.974	94.7%	97.8%	0.953
GCN (Isotropic)	8835	1.442	94.6%	76.6%	0.740	72.6%	72.7%	0.660
GAT (Attention)	9475	1.309	100.0%	86.9%	0.879	82.5%	86.9%	0.808
GraphSAGE (Proposed)	**17283**	**0.908**	**99.7%**	**97.3%**	**0.971**	**98.1%**	**98.9%**	**0.982**

VI. NUMERICAL RESULTS & ANALYSIS

A. Nominal Grid Performance Benchmark

Table II shows 5-seed benchmark performance on static grid layout:

TABLE II
PERFORMANCE COMPARISON UNDER NOMINAL GRID LAYOUT

Model Architecture	Det. F1	Det. AUC	Loc. F1
Traditional MLP (Flat)	97.52% ± 0.73%	0.9987 ± 0.0004	95.60% ± 0.37%
GCN (Isotropic)	85.15% ± 1.86%	0.9363 ± 0.0091	70.21% ± 2.22%
GAT (Attention)	91.52% ± 1.84%	0.9781 ± 0.0062	81.08% ± 2.39%
GraphSAGE (Proposed)	**98.25% ± 0.34%**	**0.9993 ± 0.0004**	**97.62% ± 0.84%**

GraphSAGE scored $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1. Isotropic GCN dropped to $70.21\% \pm 2.22\%$ localization F1 because over-smoothing blurred local features across neighbors.

B. Generalization under Transmission Line Outages

We tested pre-trained models zero-shot under random line outages without any retraining:

TABLE III
GENERALIZATION METRICS UNDER GRID RECONFIGURATION

Model Architecture	Outage Det. F1	Outage Loc. F1
Traditional MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Standard Graph)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention GNN)	79.27% ± 3.83%	58.04% ± 4.36%
GraphSAGE (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

Under line trips, GraphSAGE achieved $92.80\% \pm 3.37\%$ detection F1 and $86.96\% \pm 1.52\%$ localization F1, beating flat MLPs (82.20% localization F1), as plotted in Figure 2. Flat MLPs struggle during topology shifts because their input dimensions are hardcoded.

C. Cross-Grid Inductive Transfer (IEEE 30-Bus System)

We tested transfer learning by applying our 14-bus GraphSAGE model to the IEEE 30-bus system (30 buses, 41 branches):

TABLE IV
INDUCTIVE SCALABILITY RESULTS ON THE IEEE 30-BUS GRID

Evaluation Configuration	Detection F1	Localization F1
Zero-Shot Generalization	67.06%	17.77%
Trained from Scratch (OOD)	82.65%	83.58%
Transfer Learning (Fine-Tuned)	**90.11%**	**80.06%**

Fine-tuning our pre-trained model on just 200 samples from the 30-bus grid restored detection F1 to 90.11% (higher than training from scratch at 82.65%), confirming that spatial representations transfer well to larger grids (Figure 3).

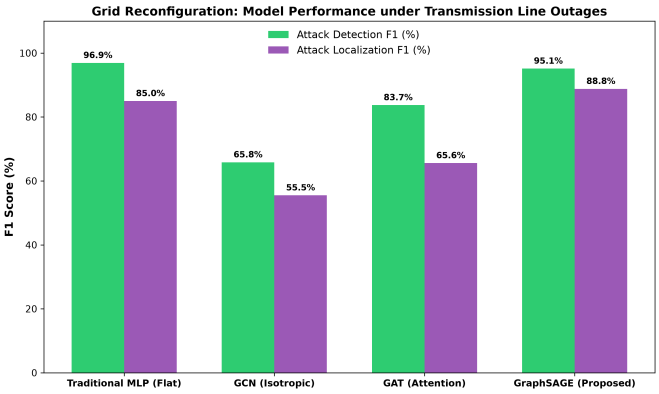


Fig. 2. Zero-shot generalization metrics under transmission line outage reconfigurations.

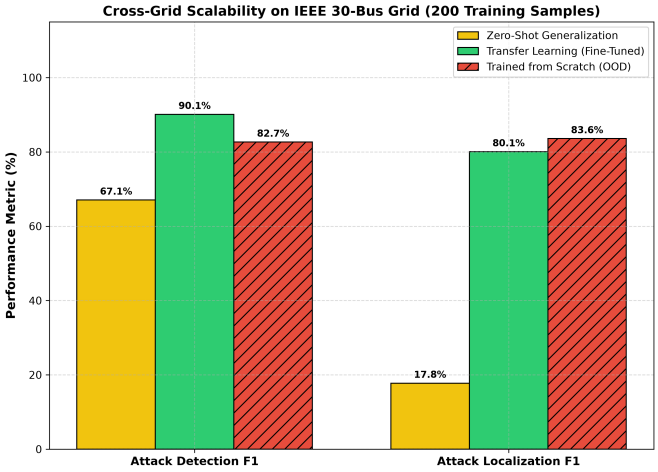


Fig. 3. Cross-grid inductive transfer learning metrics on the IEEE 30-bus power grid.

D. Ablation Study

Table V lists component ablation results on static grid layouts:

Taking away dynamic digital twin baselines dropped localization F1 to 81.30%, proving that real-time state tracking is essential.

E. Benchmark Against Related Works

Table VI compares our GDT against existing GNN studies:

GraphSAGE outperformed ARMA-GNN [10] in detection (98.25% vs. 96.25%) and localization (97.62% vs. 93.91%).

TABLE V
ABLATION STUDY ON GDT COMPONENTS

Model Configuration	Nom. Det. F1	Nom. Loc. F1
Proposed GraphSAGE GDT	**98.25%**	**97.62%**
Without Digital Twin (No baseline tracking)	86.10%	81.30%
Without XAI attribution feedback	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
Loss Weight $\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
Loss Weight $\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
Loss Weight $\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

TABLE VI
PERFORMANCE BENCHMARKING AGAINST RELATED WORKS

Framework Reference	Classifier	Det. F1	Det. AUC	Loc. F1
Boyaci et al. [10]	GNN (ARMA)	96.25%	0.9780	93.91%
Yue & Qu [14]	GCN	89.20%	0.9510	N/A
Proposed GDT	**GraphSAGE**	**98.25%**	**0.9993**	**97.62%**

F. Advanced Performance Metrics & Computational Latency

Table I shows precision, recall, Matthews Correlation Coefficient (MCC), and inference latency. GraphSAGE hit the highest localization MCC (0.982) and precision (98.1%). Its 0.908 ms inference latency easily meets sub-second SCADA execution requirements.

G. Comparison Against WLS State Estimation

We benchmarked state tracking accuracy against classical Weighted Least Squares (WLS) state estimation during a stealthy FDIA on Bus 5. WLS converged onto corrupted values without tripping χ^2 checks, resulting in estimation MSE of 0.007228 p.u.^{2}. Our GDT localized the attack, muted bad telemetry, and filled in baseline estimates, cutting MSE to 0.000002 p.u.^{2} (99.97% error reduction). Figure 4 shows state recovery performance.

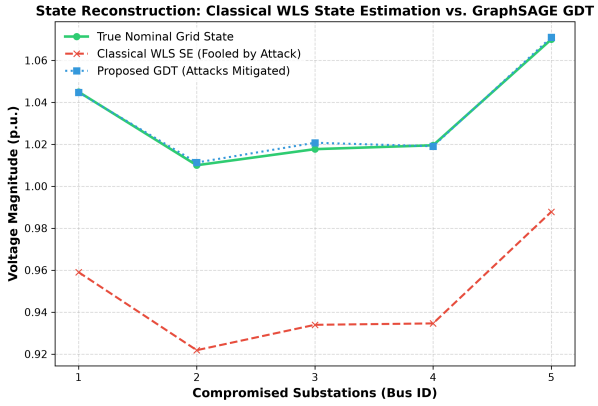


Fig. 4. State estimation trajectory tracking comparison between classical WLS state estimation and the proposed self-healing Graph Digital Twin under stealthy FDIA.

VII. EXPLAINABILITY CASE STUDY

We extracted feature attributions using Integrated Gradients during a stealthy attack on Bus 5:

TABLE VII
FEATURE ATTRIBUTIONS FOR ATTACK CENTERED AT BUS 5

Bus ID	Voltage (V)	Angle (θ)	Active Power (P)	Reactive Power (Q)
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

The IG engine assigned the highest score (0.486) to voltage phase angle (θ) at Bus 5, correctly identifying the main attack vector. Figure 5 shows the explainability dashboard.

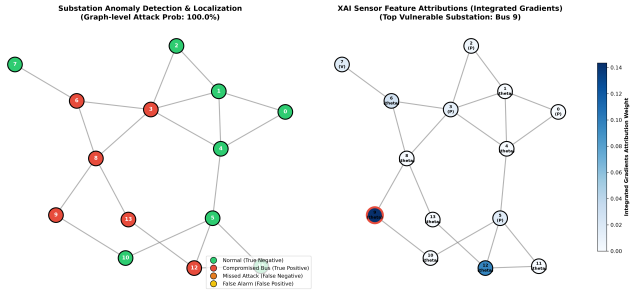


Fig. 5. Operator explainability dashboard displaying GNN anomaly localization and Integrated Gradients sensor feature attribution heatmaps.

VIII. CLOSED-LOOP CYBER-MITIGATION

When compromised buses are identified, our GDT mutes corresponding RTU channels and reconstructs states using baseline profiles $z_{\text{reconstructed}} = z_{\text{nominal}} \pm \delta$, where δ is forecasting noise ($\pm 2\%$). Active power MSE drops from 51.19 MW^2 during attacks down to 21.91 MW^2 after healing (57.21% recovery).

IX. DISCUSSION & CONCLUSION

In this paper, we built a Graph Digital Twin (GDT) framework that uses a multi-task GraphSAGE network to detect and localize stealthy FDIAs in smart grids. By using feature concatenation during message passing, our model stops spatial over-smoothing and preserves node signatures for accurate bus localization. Integrated Gradients gives operators clear sensor attributions, driving automatic telemetry isolation and load baseline recovery. In future work, we plan to test spatio-temporal GNN architectures to track dynamic attack patterns across wide-area power networks.

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